Scientific Report

Study Objective

Evaluate whether a radiomics■derived CT imaging signature can accurately identify high■risk neuroblastoma patients (predictive performance target: AUC ≥ 0.85).

Methods

Population: 13 pediatric subjects (7 neuroblastoma, 6 healthy) with baseline contrast■enhanced CT scans.

Feature Extraction: 10 top radiomic features (e.g., original_glrlm_RunVariance, original_gldm_DependenceNonUniformity, original_shape_SurfaceVolumeRatio).

Model Development: Reproducibility filtering (ICC > 0.80) → LASSO feature selection → logistic■regression classifier.

Validation: 5■fold cross■validation on the training set; independent evaluation on a 4■case validation set.

Results

| Metric | Cross■validation (mean) | Independent Set | |--------|-------------------------|-----------------| | Accuracy | 0.54 | 0.75 | | AUC | 0.37 | – (single■fold) | | Recall (Sensitivity) | 0.54 | 0.75 | | Precision (PPV) | 0.39 | 0.67 | | F1■Score | 0.44 | 0.75 |

One healthy case was mis■classified as neuroblastoma in the independent set.

The overall AUC (0.37) is far below the predefined success threshold.

Conclusion

The radiomics signature demonstrated insufficient discriminative ability to predict high■risk neuroblastoma. The hypothesis is rejected.

Recommendations

Increase sample size (≥200 subjects) to improve model training and stability.

Harmonize imaging protocols across centers and explore additional modalities (e.g., MRI, PET).

Consider ensemble learning or deep■learning approaches after adequate data collection.

Report generated by the scientific reporting assistant.